Line Parameters IV

Calculate one verified R-X-B record from conductor and tower data

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Lecture outline

Prepare the geometry once, calculate the phase parameters, and assemble one complete R, X, B record per physical circuit.

1. Geometry: identify circuits and bundles; calculate phase GMD and both bundle self-distances.
2. Parameters: calculate resistance, inductance, and capacitance rates; obtain X' and B' at frequency f.
3. Route record: form physical totals, charging quantities, and per-unit values.
4. Exercises: follow the same geometry-to-parameters sequence and verify the record.

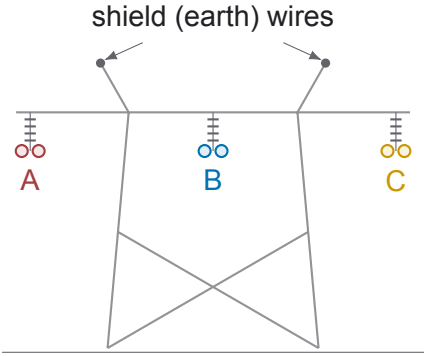
Textbook: Glover, Overbye, and Sarma, *Power System Analysis and Design*, 6th ed. (2017). Sections 4.4, 4.6, 4.10, and 4.13. Detailed references are retained in the instructor notes.

PART 1

Geometry and equivalent distances

Identify the circuits and phase bundles, then prepare the phase GMD and the magnetic and electric self-distances needed for the parameter calculations.

Reading a transmission tower



schematic front view; not to scale

Tower and crossarms
Support the conductors and maintain clearances.

Insulator strings
Support and electrically isolate the phase conductors.

Shield wires
Provide lightning protection; they are not extra phases.

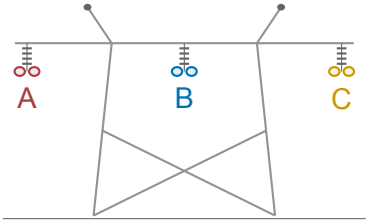
Each pair: one phase, two subconductors.

A, B, and C are the three phases; each illustrated phase contains two subconductors.

One tower can carry two circuits

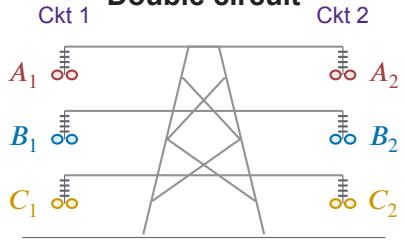
One three-phase circuit contains one A–B–C set. Each phase may be bundled.

Single circuit



$3 \times 2 = 6$ phase conductors

Double circuit

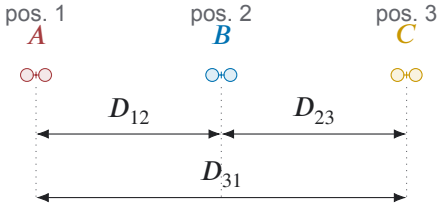


$2 \times 3 \times 2 = 12$ phase conductors

Illustrative layouts; shield wires excluded from counts and omitted on the right.

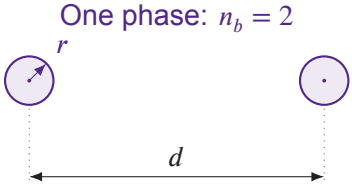
The physical distances

Between phases



D_{12}, D_{23}, D_{31}: spacings between fixed phase-bundle positions 1, 2, 3.

Within one phase



n_b: subconductors per phase.
r: outside radius of one member.
d: member-center spacing.

Small-bundle approximation: $r \ll d \ll$ phase spacing. Diagrams use different scales.

Where equivalent distances enter

Balanced, completely transposed line in air; earth and shield-wire effects neglected.

Inductance

$$L'_\phi = 2 \times 10^{-7} \ln \frac{D_m}{D_{s\mathrm{L}}} \quad \text{[H/m]}$$

Capacitance

$$C'_n = \frac{2\pi\epsilon_0}{\ln(D_m/D_{s\mathrm{C}})} \quad \text{[F/m]}$$

D_m Mutual geometry: one equivalent spacing between phases.

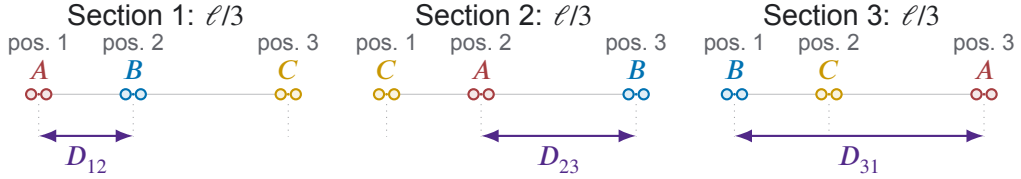
$D_{s\mathrm{L}}$ Magnetic self-distance: one equivalent for a whole phase bundle.

$D_{s\mathrm{C}}$ Electric self-distance: a different equivalent for the same bundle.

Prepare one mutual distance and two self-distances before calculating the parameters.

Geometric mean distance (GMD)

Follow A and B as phase bundles change positions along the route.



Keep $D_{s\mathsf{L}}$ fixed; choose D_m to match the averaged inductive effect:

$$\ln\frac{D_m}{D_{s\mathsf{L}}} = \frac{1}{3}\left(\ln\frac{D_{12}}{D_{s\mathsf{L}}} + \ln\frac{D_{23}}{D_{s\mathsf{L}}} + \ln\frac{D_{31}}{D_{s\mathsf{L}}}\right)$$

$$= \ln\frac{(D_{12}D_{23}D_{31})^{1/3}}{D_{s\mathsf{L}}} \quad \Longrightarrow \quad \boxed{D_m = (D_{12}D_{23}D_{31})^{1/3}}.$$

One GMD for inductance and capacitance

Electric potential has the same logarithmic dependence on distance as magnetic flux linkage.

The same phase spacings and transposition average therefore give **the same D_m for L and C**.

The distinction lies in the **self terms** of the conductors.

The self term of one subconductor

Use s as a common symbol for one subconductor's magnetic or electric self-distance.

Magnetic self term (for L)



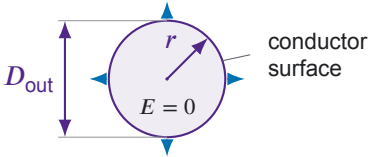
D_s: Geometric Mean Radius (GMR)

Solid round, nonmagnetic; uniform current.

$$s = D_s = re^{-1/4} \approx 0.7788r$$

Stranded conductors: use supplied GMR.

Electric self term



D_{out}: physical outside diameter

r: physical outside radius of one member.

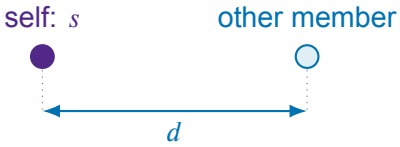
$$s = r = \frac{D_{\text{out}}}{2} \quad \text{for } C$$

Potential is evaluated at the surface.

D_s belongs to one member; $D_{s\text{L}}$ will describe the whole phase bundle.

Two-member bundle self-distances

Identical symmetric members; bundle size is small compared with phase spacing.



Equal sharing gives two weights of $1/2$:

$$I_1 = I_2 = \frac{I_\phi}{2}, \qquad q_1' = q_2' = \frac{q_\phi'}{2}.$$

$$\frac{1}{2}\left[\underbrace{\ln\frac{D_m}{s}}_{\text{self}} + \underbrace{\ln\frac{D_m}{d}}_{\text{other member}}\right] = \ln\frac{D_m}{\sqrt{sd}} \quad \Longrightarrow \quad s_{\text{eq}} = \sqrt{sd}.$$

Magnetic bundle: $s = D_s$

$$\boxed{D_{s\text{L}} = \sqrt{D_s d}}$$

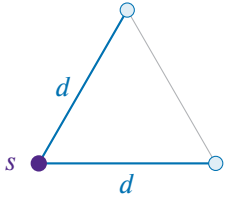
Electric bundle: $s = r$

$$\boxed{D_{s\text{C}} = \sqrt{rd}}$$

The distance products for common bundles

With equal sharing, start at the purple member: its self term is s.

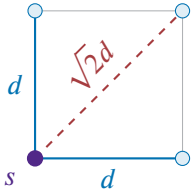
Three members: equilateral



Three factors: $s,\ d,\ d$.

$$s_{\text{eq}} = (sd^2)^{1/3}$$

Four members: square



Four factors: $s,\ d,\ d,\ \sqrt{2}d$.

$$s_{\text{eq}} = (\sqrt{2}sd^3)^{1/4}$$

Each member gives the same set of factors; their geometric mean is the bundle self-distance.

Prepared distances for the phase formulas

Phase GMD, common to L and C: $D_m = (D_{12}D_{23}D_{31})^{1/3}$

One phase	Magnetic $D_{s\mathrm{L}}$	Electric $D_{s\mathrm{C}}$
Single member	D_s	r
Two members	$(D_s d)^{1/2}$	$(rd)^{1/2}$
Three (equilateral)	$(D_s d^2)^{1/3}$	$(rd^2)^{1/3}$
Four (square)	$(\sqrt{2}D_s d^3)^{1/4}$	$(\sqrt{2}rd^3)^{1/4}$

Remaining data: member AC resistance rate R'_{sub} at a stated temperature, member count n_b, frequency f, and route length ℓ.

The geometry is complete. Calculate R, then use these prepared distances for L and C.

PART 2

Resistance, inductance, and capacitance

The geometric inputs are ready. Use current sharing for resistance and the prepared self-distances in the established inductance and capacitance formulas.

Rates, totals, and per unit

Per length		Physical total		Per unit
R'_ϕ	$\xrightarrow{\times \ell}$	R_ϕ	$\xrightarrow{\div Z_{\text{base}}}$	r_{pu}
Ω/km		Ω		p.u.
One unit length of a uniform phase		One phase over the full route		Full-route total on a declared base

Unit checks

Logarithms

Distances must use the same unit.

$\text{m}/\text{m} = 1$

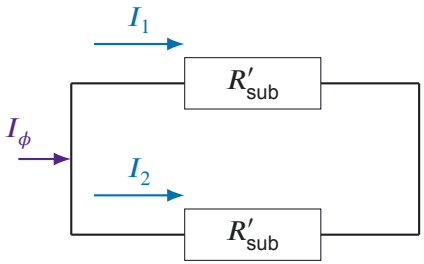
Route scaling

Use matching rate and route units.

$(\Omega/\text{km}) \times \text{km} = \Omega$

Resistance of a two-member bundle

Identical symmetric members share the same terminal voltage.



resistive model of a unit-length slice

$$I_\phi = I_1 + I_2, \qquad I_1 = I_2 = I_\phi/2.$$

The common resistive drop per length is

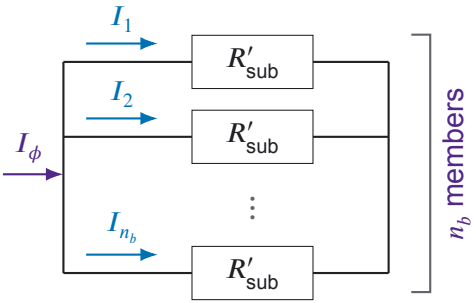
$$R'_{\text{sub}} \frac{I_\phi}{2} = R'_\phi I_\phi.$$

$$\boxed{R'_\phi = R'_{\text{sub}}/2}$$

Two identical, equally sharing members halve the phase resistance rate and the resistive drop at fixed phase current.

Resistance of a general bundle

A phase bundle contains n_b identical, equally sharing subconductors.



one phase; unit-length resistive model

Total phase current

$$I_\phi = \sum_{k=1}^{n_b} I_k = n_b I_k.$$

Common resistive drop per length

$$R'_\phi I_\phi = R'_{\text{sub}} \frac{I_\phi}{n_b}.$$

$$\boxed{R'_\phi = \frac{R'_{\text{sub}}}{n_b}}$$

Equal sharing gives $R'_\phi = R'_{\text{sub}}/n_b$ for n_b identical members.

Inductance and capacitance rates

Magnetic inputs: D_m, $D_{s\mathsf{L}}$

$$L'_\phi = 2\times 10^{-7} \ln \frac{D_m}{D_{s\mathsf{L}}}$$

[H/m]

At angular frequency $\omega = 2\pi f$:

$$X'_\phi = \omega L'_\phi \qquad [\Omega/\text{m}]$$

Electric inputs: D_m, $D_{s\mathsf{C}}$

$$C'_n = \frac{2\pi\epsilon_0}{\ln(D_m/D_{s\mathsf{C}})}$$

[F/m]

At angular frequency $\omega = 2\pi f$:

$$B'_{\text{sh}} = \omega C'_n \qquad [\text{S/m}]$$

Air: $\epsilon_0 \approx 8.85419\times 10^{-12}$ F/m. Distances within each logarithm use the same unit.

At fixed D_m and f, larger bundle self-distances give $L'\downarrow$, $X'\downarrow$, $C'\uparrow$, and $B'\uparrow$.

PART 3

Route totals and per unit

Resistance, reactance, and susceptance rates are ready. Apply route length once, then form charging quantities and the record on declared system bases.

Physical route totals

Series parameters of one phase

$$R_\phi = R'_\phi \ell, \qquad X_\phi = X'_\phi \ell$$

$$Z = R_\phi + \mathrm{j} X_\phi \quad [\Omega]$$

Shunt parameters of one phase

$$C_n = C'_n \ell, \qquad B_{\mathrm{sh}} = B'_{\mathrm{sh}} \ell$$

$$Y = \mathrm{j} B_{\mathrm{sh}} \quad [\mathrm{S}]$$

These are full-route totals for one phase of one physical circuit. They contain the bundle effect already.

The route length enters once; the resulting Z and Y describe one phase of one circuit.

Per-unit line record

The physical totals are ready. To join equipment from the same voltage zone in one network model, choose line-to-line voltage base $V_{LL,\text{base}}$ and total three-phase power base S_{base}.

$$Z_{\text{base}} = \frac{V_{LL,\text{base}}^2}{S_{\text{base}}}, \qquad Y_{\text{base}} = \frac{1}{Z_{\text{base}}}$$

Series normalization

$$r_{\text{pu}} = \frac{R_\phi}{Z_{\text{base}}}, \qquad x_{\text{pu}} = \frac{X_\phi}{Z_{\text{base}}}$$

Shunt normalization

$$b_{\text{pu}} = \frac{B_{\text{sh}}}{Y_{\text{base}}} = B_{\text{sh}} Z_{\text{base}}$$

Per-unit values are dimensionless ratios of full-route totals to their declared bases.

Charging quantities

For a charging estimate at uniform line-to-line RMS voltage V_{LL},

$$V_\phi = \frac{V_{LL}}{\sqrt{3}}, \qquad |I_{\text{ch},\phi}| = B_{\text{sh}} V_\phi$$

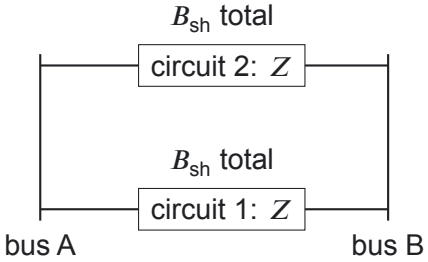
$$Q_C = 3 B_{\text{sh}} V_\phi^2 = B_{\text{sh}} V_{LL}^2$$

$Q_C > 0$ is capacitive reactive power supplied. Signed absorbed power is $-Q_C$.

A nominal-π places $B_{\text{sh}}/2$ at each end. Different terminal voltages require separate endpoint calculations.

Charging uses total susceptance and operating voltage. The line's MVA limit remains separately supplied data.

Two parallel circuits



Identical circuits connect the same buses; mutual coupling is neglected.

$$Z_{eq} = \left(\frac{1}{Z} + \frac{1}{Z}\right)^{-1} = Z/2.$$

$$B_{eq} = B_{sh} + B_{sh} = 2B_{sh}.$$

Each circuit's Z, B_{sh} already includes its bundle. Keep both physical circuit records.

After one circuit opens, the remaining circuit has Z and B_{sh}; its bundle configuration is unchanged.

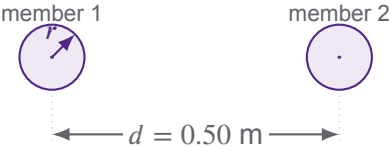
PART 4

Exercises

Prepare the distances first, calculate the parameter rates next, and finish with physical totals and per-unit records. Keep the physical circuit identity attached.

Exercise 1: given and find

One phase; two identical members



Cross-section; not to scale

Given data

$$D_s = 0.0100 \text{ m},$$
$$r = 0.0120 \text{ m},$$
$$R'_{\text{sub}} = 0.0600 \ \Omega/\text{km},$$
$$\ell = 100 \text{ km}.$$

D_s, r, and R'_{sub} describe one member.

1. Prepare the magnetic and electric bundle self-distances.
2. Find the phase resistance rate and the total resistance over the 100-km route.
3. For two identical uncoupled circuits in parallel, state the equivalent Z and B.

Exercise 1: solution

Prepared self-distances

$$D_{s\mathrm{L}} = \sqrt{0.0100(0.50)} \approx 0.07071 \text{ m}, \qquad D_{s\mathrm{C}} = \sqrt{0.0120(0.50)} \approx 0.07746 \text{ m}$$

Resistance rate and route total

$$R'_{\phi} = \frac{0.0600}{2} = 0.0300 \ \Omega/\mathrm{km}, \qquad R_{\phi} = 0.0300(100) = 3.00 \ \Omega$$

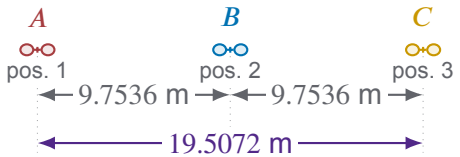
Two identical uncoupled circuits

$$Z_{\text{parallel}} = Z_{\text{circuit}}/2, \qquad B_{\text{parallel}} = 2B_{\text{circuit}}$$

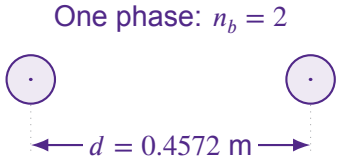
Member geometry and current sharing describe one phase; circuit parallelism combines complete circuit records.

Exercise 2: given

Balanced, fully transposed, one physical circuit; two members per phase.



Phase-bundle centers; one transposition section



Member-center spacing; enlarged schematic

$V_{LL} = V_{LL,\text{base}}$	500.0 kV
f	60.0 Hz
ℓ	128.74752 km
S_{base}	100.0 MVA

$R'_{\text{sub,ac,25°C}}$	0.0344488189 Ω/km
GMR D_s	0.01792224 m
Diameter D_{out}	44.7548 mm

All conductor data describe one member.

Use phase-bundle centers since $d \ll 9.7536$ m. Diagram scales differ.

Supplied planning limit: 1000 MVA/circuit, independent of R, X, and B.

Exercise 2: find

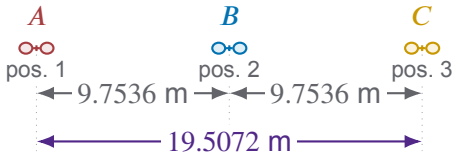
Find and verify

(a) Prepare D_m, the member radius r, and the bundle self-distances $D_{s\mathrm{L}}$ and $D_{s\mathrm{C}}$.
(b) Calculate R'_ϕ, then L'_ϕ, X'_ϕ, C'_n, and B'_{sh} using the prepared geometry.
(c) Form the full-route physical totals, charging current, and supplied Mvar.
(d) Convert to the 500-kV, 100-MVA bases. Compare one member with the two-member bundle and verify the result.

A usable solution includes units, model assumptions, and the declared per-unit bases.

Ex. 2(a): phase GMD

Small-bundle phase-center approximation



Phase-bundle centers; one transposition section

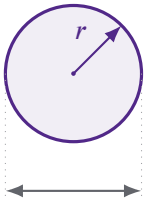
$$
\begin{aligned}
D_m &= (D_{12} D_{23} D_{31})^{1/3} \\
&= (9.7536^2 \cdot 19.5072)^{1/3} \\
&\approx \boxed{12.2888 \text{ m}}.
\end{aligned}
$$

Since $d = 0.4572$ m $\ll 9.7536$ m, each mutual bundle distance is represented by its phase-center spacing.

Complete transposition replaces the three phase spacings by their geometric mean.

Ex. 2(a): member radius

One subconductor



$D_{\text{out}} = 44.7548$ mm

Enlarged cross-section

The supplied outside dimension is a diameter. The electric self term uses the radius:

$$\begin{aligned} r &= \frac{D_{\text{out}}}{2} \\ &= \frac{44.7548}{2} \text{ mm} \\ &= 22.3774 \text{ mm} \\ &= 0.0223774 \text{ m}. \end{aligned}$$

Physical radius and supplied magnetic GMR are separate member inputs for the two bundle self-distances.

Ex. 2(a): bundle self-distances

Prepared phase GMD: $D_m \approx 12.2888$ m; member spacing: $d = 0.4572$ m.

Magnetic self-distance

$$\begin{aligned} D_{s\mathrm{L}} &= \sqrt{D_s d} \\ &= \sqrt{(0.01792224)(0.4572)} \\ &\approx \boxed{0.0905210 \text{ m}}. \end{aligned}$$

Electric self-distance

$$\begin{aligned} D_{s\mathrm{C}} &= \sqrt{rd} \\ &= \sqrt{(0.0223774)(0.4572)} \\ &\approx \boxed{0.101148 \text{ m}}. \end{aligned}$$

Both ratios are dimensionless; all three distances use metres.

$$\ln \frac{D_m}{D_{s\mathrm{L}}} \approx 4.91085912 \qquad \ln \frac{D_m}{D_{s\mathrm{C}}} \approx 4.79985456$$

Geometry is complete: the magnetic ratio enters L', and the electric ratio enters C'.

Ex. 2(b): resistance rate

Supplied 25°C AC rate

One member:

$$R'_{\text{sub}} = 0.0344488189\ \Omega/\text{km}.$$

Two identical members share the phase current equally:

$$\begin{aligned} R'_{\phi} &= \frac{R'_{\text{sub}}}{2} \\ &= \frac{0.0344488189}{2}\ \Omega/\text{km} \\ &\approx \boxed{0.0172244094\ \Omega/\text{km}}. \end{aligned}$$

The derived bundle rate retains the supplied 25°C AC condition.

> **Two identical members halve the phase resistance rate. The number of phases never multiplies this rate.**

Ex. 2(b): inductance and reactance

$$L'_\phi \approx 2 \times 10^{-7}(4.91085912) \approx 9.82171824 \times 10^{-7} \text{ H/m}.$$

Express the rate per kilometre:

$$L'_\phi \approx 0.982171824 \text{ mH/km}.$$

At 60 Hz,

$$\boxed{X'_\phi \approx 2\pi(60)(0.982171824 \times 10^{-3}) \approx 0.370270054 \ \Omega\text{/km}.}$$

The reactance rate is a balanced per-phase quantity; use kilometres for the route length.

Ex. 2(b): capacitance and B rate

$$C'_n \approx \frac{2\pi(8.8541878 \times 10^{-12})}{4.79985456} \text{ F/m}$$
$$\approx 1.15904559 \times 10^{-11} \text{ F/m}$$
$$\approx 11.5904559 \text{ nF/km}.$$

At 60 Hz,

$$B'_{\text{sh}} \approx 2\pi(60)(11.5904559 \times 10^{-9}) \text{ S/km}$$
$$\approx 4.36949895 \ \mu\text{S/km}.$$

All parameter rates now use kilometres; multiply each by the same route length.

Ex. 2(c): physical totals

Route length: $\ell = 128.74752$ km. Multiply every per-kilometre rate by ℓ.

Quantity	Per-kilometre rate	One-circuit route total
R_ϕ	0.0172244 Ω/km	2.2176 Ω
X_ϕ	0.370270 Ω/km	47.6714 Ω
C_n	11.5905 nF/km	1.49224 μF
B_{sh}	4.36950 μS/km	0.562562 mS

$$Z = 2.2176 + \mathrm{j}\,47.6714\ \Omega, \qquad Y = +\mathrm{j}\,0.562562\ \text{mS}.$$

Every total belongs to one phase of one physical circuit over the full route.

Ex. 2(c): charging quantities

The balanced shunt uses phase-to-neutral voltage:

$$V_{\phi,\mathrm{rms}} = \frac{500\ \mathrm{kV}}{\sqrt{3}} \approx 288.675135\ \mathrm{kV}.$$

$$|I_{\mathrm{ch},\phi}| = B_{\mathrm{sh}} V_{\phi,\mathrm{rms}} \approx \boxed{162.398\ \mathrm{A}}.$$

The three-phase charging magnitude closes directly:

$$\boxed{|Q_{\mathrm{ch},3\phi}| = B_{\mathrm{sh}} V_{LL,\mathrm{rms}}^2 \approx 140.640538\ \mathrm{Mvar}.}$$

The line rating remains a supplied planning limit. Charging Mvar is a separate electrical result from B_{sh} and voltage.

Ex. 2(d): per-unit record

Declared bases

$$Z_{\text{base}} = \frac{(500\ \text{kV})^2}{100\ \text{MVA}} = 2500\ \Omega,$$

$$Y_{\text{base}} = 0.0004\ \text{S} = 0.400\ \text{mS}.$$

Course worksheet record

$$\boxed{\begin{aligned} r_{\text{pu}} &= 0.00088704, \\ x_{\text{pu}} &= 0.0190685, \\ b_{\text{pu}} &= 1.40641. \end{aligned}}$$

b_{pu} is the total one-circuit shunt-susceptance magnitude. The supplied planning limit is 1000 MVA.

The physical line-data record is complete. Software-specific mapping remains a later task.

Ex. 2(d): bundle comparison

Parameter comparison under one fixed geometry

The phase centers, subconductor type, route, frequency, and model remain fixed while bundle count changes. Material-quantity and rating comparisons require a separate design study.

Rate	One member	Two-member bundle	Change
R'_ϕ [Ω/km]	0.0344488	0.0172244	−50.000%
X'_ϕ [Ω/km]	0.492380	0.370270	−24.800%
C'_n [nF/km]	8.81881	11.5905	+31.429%

Under the declared assumptions, bundling lowers resistance and reactance while increasing capacitance and charging.

Ex. 2(d): verification

Check	Expected signature
Self quantities	$D_{s\mathrm{L}}$ appears in X, and $D_{s\mathrm{C}}$ appears in C.
Route-length closure	Per-kilometre rate times 128.74752 km returns each unprimed total exactly once.
Charging closure	$B_{\mathrm{sh}} V_{LL,\mathrm{rms}}^2 = 140.641$ Mvar.
Per-unit closure	$b_{\mathrm{pu}} S_{\mathrm{base}} = 140.641$ Mvar at 1.0 p.u.
Circuit identity	Each physical circuit keeps a full row, ID, and separately supplied limit.

A correct answer must close numerically, dimensionally, physically, and topologically.

Summary

- **Prepared geometry:** phase GMD describes mutual spacing. Magnetic GMR and physical radius give the two bundle self-distances.
- **Parameter calculation:** current sharing gives resistance; the prepared distances enter L and C. Frequency gives X and B, then route length gives totals.
- **Per-unit conversion:** convert the complete physical row with the correct system power base and voltage-zone base.
- **Parallel circuits:** retain separate rows for separate physical circuits. The simple $Z/2$, $2B$ equivalent assumes identical uncoupled circuits; nearby circuits can require mutual terms.

A reliable branch model preserves both its electromagnetic data and its topological identity.